

Respiration in Plants

1. Which of the following complex in ETS during aerobic respiration is not involved in the electron transfer ?

- (1) Complex I and II
- (2) Complex II and III
- (3) Complex IV
- (4) Complex V

2. Which of the following product is formed by Krebs cycle ?

- (1) NADPH + H⁺
- (2) ADP
- (3) ATP
- (4) NADP⁺

3. Mark the correct statement

- (1) Fats are preferred energy fuel
- (2) Glycerol enters glycolysis through serine formation
- (3) In cellular respiratory ETS, electron movement releases energy
- (4) A number of TCA cycle intermediates are used in various catabolic reactions only.

4. How many ATP will be produced from two molecules of succinyl CoA till it gets catabolised into oxaloacetic acid through oxidative phosphorylation only

- (1) 10
- (2) 11
- (3) 12
- (4) 8

5. Read the following statement from i – iv and identify the correct ones

- (i) Yeast poison themselves to death when the concentration of alcohol reaches about 13%

(ii) In fermentation less than 7% of the energy in glucose is released.

(iii) Net gain in both types of fermentation is 4 ATP (with respect to one glucose molecule)

(iv) During glycolysis enzyme involved in phosphorylation of hexose sugar is Enolase

- (1) i, ii, iii
- (2) ii, iii, iv
- (3) i and ii
- (4) ii and iii

6. Mark the incorrect statement

- (1) Cytochrome C is a peripheral protein
- (2) RQ of maturing fatty seeds is > 1
- (3) Succinyl CoA is an intermediate of TCA cycle
- (4) In glycolysis, during oxidation electrons are removed by molecular oxygen

7. Read the following statements and state True (T) or False (F)

- A. Oligomycin inhibits oxidative phosphorylation
- B. During aerobic respiration, end products are lactic acid or alcohol
- C. The RQ value of oxalic acid is 4.
- D. Cytochrome-c is a non-mobile protein attached to inner membrane

Select the correct option

- | | A | B | C | D |
|-----|---|---|---|---|
| (1) | T | F | T | F |
| (2) | T | T | T | F |
| (3) | T | F | F | T |
| (4) | T | T | F | T |

8. Site of link reaction is

- (1) Mitochondrial matrix
- (2) Cytoplasm
- (3) Perimitochondrial space
- (4) Both (1) & (3)

9. How many ATP will be formed through substrate level phosphorylation in Krebs' cycle for one molecules of sucrose ?

- (1) 1
- (2) 4
- (3) 2
- (4) 6

10. In aerobic respiration

(i) Oxygen acts as terminal electron acceptor

(ii) Transfer of electrons from Complex III to Complex IV occurs by cytochrome oxidase

(iii) Ubiquinone also receives reducing equivalent via FADH₂

(iv) Krebs' cycle doesn't involve any substrate level phosphorylation

Which of the above statements are true ?

- (1) i, ii, iv
- (2) ii, iii, iv
- (3) i and iii
- (4) i and ii

11. The term floating respiration is used for the oxidation of _____

- (1) Carbohydrate and Fat
- (2) Organic acid
- (3) Fat and organic acid
- (4) Protein and Fat

12. How many enzyme complex/complexes is/are involved in oxidative phosphorylation ?

- (1) Four
- (2) Five
- (3) One
- (4) Two

13. Find the incorrect match from the following

- (1) Glycolysis – Cytosol
- (2) β -oxidation of fatty acids – Mitochondrial matrix
- (3) ETC – Inner mitochondrial membrane
- (4) TCA cycle – Ribosome

14. The metallic ions that are required in the first step and last step of glycolytic pathway are A and B and these are activator molecule of enzyme C. A, B and C respectively represents

- (1) Mn^{2+} , K^+ and Enolase
- (2) Mg^{2+} , K^+ and Pyruvate kinase
- (3) Ca^{2+} , Mg^{2+} and Kinase
- (4) Mn^{2+} , Mg^{2+} and Mutase

15. Find the correct statement from the following

- (1) Alcoholic fermentation of pyruvate requires only alcohol dehydrogenase
- (2) Lactobacillus lacti is a heterofermentative lactic acid bacteria
- (3) Less than seven percent of the energy in glucose is released during fermentation
- (4) CO_2 is released during lactic acid fermentation

16. Read the statement carefully

A. Three $NADH + H^+$ and one $FADH_2$ molecules are generated during two turns of TCA cycle

B. Two molecules of CO_2 are released per molecule of acetyl CoA oxidized in TCA cycle

C. The continued oxidation of acetyl CoA via the TCA cycle requires the continued replenishment of OAA.

D. Krebs' cycle is an amphibolic pathway

How many of the above statements are correct?

- (1) All are correct except A
- (2) A, B and D are correct
- (3) A and B are incorrect
- (4) A, B, C and D are correct

17. What is not correct with respect to ETC?

- (1) Ubiquinone receives electrons from $\text{NADH} + \text{H}^+$ and FADH_2
- (2) The reduced ubiquinone is called ubiquinol
- (3) Cytochrome c, a and a_3 is iron containing enzyme
- (4) O_2 is terminal electron acceptor generating metabolic water

18. Match the column correctly

Column - I

A. Acetyl CoA

B. Succinyl CoA

C. Oxaloacetic acid

D. α -keto glutaric acid

Column - II

(i) Precursor for amino acid synthesis

(ii) Raw material for alkaloids, pyrimidines

(iii) Raw material for cytochrome, chlorophyll

(iv) Raw materials for carotenoids and gibberellins

Options:

- (1) A – (iii), B – (ii), C – (i), D – (iv)
- (2) A – (i), B – (iii), C – (iv), D – (ii)
- (3) A – (iv), B – (iii), C – (ii), D – (i)
- (4) A – (iv), B – (iii), C – (i), D – (ii)

19. The total number of ATP molecules produced through substrate level phosphorylation in Krebs' cycle for one molecule of sucrose is

- (1) 2 ATP
- (2) 4 ATP
- (3) 8 ATP
- (4) 48 ATP

20. How many decarboxylation reactions take place in aerobic respiration?

- (1) 3
- (2) 2
- (3) 1
- (4) 0

21. Which of the following organisms can prepare their own food by the process of photosynthesis?

- (1) Herbivores
- (2) Saprophytes
- (3) Cyanobacteria
- (4) Heterotrophs

22. In anaerobic organisms, the only process in respiration is

- (1) EMP pathway
- (2) Tricarboxylic acid cycle
- (3) Krebs' cycle
- (4) Citric acid cycle

23. Sucrose is converted into glucose and fructose by the help of the enzyme

- (1) Amylase
- (2) Invertase
- (3) Maltase
- (4) Zymase

24. In glycolysis, a chain of ten reactions, under the control of different enzymes, takes place to produce

- (1) Sucrose from glucose
- (2) Glucose from fructose
- (3) Pyruvate from acetyl CoA
- (4) Pyruvate from glucose

25. During glycolysis ATP is utilised in the conversion of

- (1) Glucose into glucose-6-phosphate
- (2) Glucose-6-phosphate into fructose-6-phosphate
- (3) Phosphoenol pyruvate into pyruvic acid
- (4) 3-phosphoglyceric acid into 2-phosphoglycerate

26. The key product of glycolysis is

- (1) DHAP
- (2) Pyruvic acid
- (3) Phosphoenolpyruvate
- (4) Acetyl CoA

27. For complete oxidation of glucose to CO₂ and H₂O organisms adopt

- (1) Lactic acid fermentation
- (2) Alcoholic fermentation
- (3) Krebs' cycle
- (4) Substrate-level phosphorylation

28. Yeasts poison themselves to death when the concentration of alcohol reaches about

- (1) 5 percent
- (2) < 5 percent
- (3) < 8 percent
- (4) 13 percent

29. The enzyme which catalyses the conversion of pyruvate into acetyl CoA in the mitochondrial matrix is

- (1) Lactate dehydrogenase
- (2) Succinate dehydrogenase
- (3) Pyruvate dehydrogenase
- (4) Pyruvate kinase

30. Substrate level phosphorylation in the TCA-cycle takes place during the conversion of

- (1) Succinyl-CoA to succinic acid
- (2) Oxaloacetic acid to citric acid
- (3) Malic acid to oxaloacetic acid
- (4) Citric acid to isocitric acid

31. Which of the following complex refers to cytochrome-c oxidase complex containing cytochromes a and a₃ and two copper centres?

- (1) Complex-V
- (2) Complex-IV
- (3) Complex-II
- (4) Complex-I

32. What will be the respiratory quotient value (RQ) for the fatty acid, tripalmitin?

- (1) > 1
- (2) 0.9
- (3) 4.0
- (4) 0.7

33. If fatty acids were to be utilised in the respiratory pathway, they would first be degraded to

- (1) Oxaloacetate
- (2) Pyruvic acid
- (3) Acetyl CoA
- (4) PGAL

34. The respiratory pathway should be considered as

- (1) Amphibolic
- (2) Anabolic only
- (3) Catabolic only
- (4) Synthetic

35. What will be the net gain of ATP molecules for each molecule of glucose used in fermentation?

- (1) 36 ATP
- (2) 38 ATP
- (3) 7 ATP
- (4) Only 2 ATP

36. The peripheral membrane protein complex, which contains the site for synthesis of ATP from ADP and inorganic phosphate, present on the inner mitochondrial membrane is called

- (1) F₁ component
- (2) F₀ component
- (3) Ubiquinone
- (4) Cytochrome-c

37. The complex-I of the electron transport system (ETS) present on the inner-mitochondrial membrane is known as:

- (1) Succinate dehydrogenase complex
- (2) NADH dehydrogenase complex
- (3) Cytochrome bc₁ complex
- (4) Cytochrome-c oxidase

38. Substrate level phosphorylation in glycolysis takes place during conversion of

- (1) 2-phosphoglycerate to 3-PGA
- (2) 2-phosphoglycerate to PEP

- (3) PEP to pyruvic acid
(4) Glucose-6-phosphate to fructose-6-phosphate

39. Glycerol would usually enter the respiratory pathway after being converted to

- (1) Fructose
(2) PGAL
(3) Amino acid
(4) Proteases

40. The small protein attached to the outer surface of the inner mitochondrial membrane and acts as a mobile carrier for transfer of electrons between complex III and IV in the electron transport system (ETS) is

- (1) Ubiquinone
(2) Ferredoxin
(3) Cytochrome-c
(4) F1 – particle

41. How many FADH_2 molecules will be produced from two molecules of glucose in aerobic respiration?

- (1) 4
(2) 2
(3) 1
(4) 8

42. Which of the following respiratory complex does not act as an electron carrier?

- (1) FADH_2
(2) NADH dehydrogenase
(3) $\text{F}_0\text{--F}_1$
(4) Cytochrome b--c_1

43. During respiration, mobile carrier Cytochrome c transfers electron in between

- (1) NADH to FADH_2
(2) Cytochrome b_1 to Cytochrome f
(3) Cytochrome b--c_1 to Cytochrome c oxidase
(4) FADH_2 to Ubiquinone

44. How many ATP will be produced from one molecule of 3-phosphoglyceraldehyde through substrate level during aerobic respiration?

- (1) Six
(2) Three
(3) Seventeen
(4) Fifteen

45. Number of ATP produced by two molecules of α -ketoglutaric acid (till its conversion into oxaloacetic acid) through oxidative phosphorylation only, is

- (1) 13
(2) 9
(3) 18
(4) 16

46. The ultimate acceptor of e^- and H^+ in cellular respiration is

- (1) Water
(2) Cytochrome a--a_3
(3) Oxygen
(4) UQ

47. RQ value of organic acid is

- (1) Equal to one
(2) More than one
(3) Less than one
(4) Equal to RQ of protein

48. Read the following statements:

- (a) RQ of maturing fatty seeds > 1
(b) 2,4-DNP is an uncoupling agent in ETS
(c) Cytochrome-c is an intrinsic protein completely embedded in the inner mitochondrial membrane
(d) Organic acids are not used as a respiratory substrate

Mark True (T) and False (F): for statements (a),(b),(c) & (d) respectively

- (1) T, T, F, F
(2) F, F, T, F
(3) T, F, T, F
(4) T, T, F, T

49. Mark the correct match:

- (1) Krebs' cycle – TCA cycle
- (2) Glycolysis – CO₂ is evolved
- (3) ETS – Outer side of inner mitochondrial membrane
- (4) Oxidative phosphorylation – Uses plastoquinone to create gradient

50. α -Ketoglutarate dehydrogenase complex catalyzes the _____ step in TCA cycle

- (1) Decarboxylation
- (2) Oxidation
- (3) Oxidative decarboxylation
- (4) Dehydration



Answer Key

1. (4)
2. (3)
3. (3)
4. (1)
5. (3)
6. (4)
7. (1)
8. (1)
9. (2)
10. (3)
11. (1)
12. (3)
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39. (2)
40. (3)
41. (1)



- 42. (3)
- 43. (3)
- 44. (2)
- 45. (4)
- 46. (3)
- 47. (2)
- 48. (1)
- 49. (1)
- 50. (3)

